Empirical Evaluation and Architectural Boundaries of Small Language Models (SLMs) under Severe Edge Memory Constraints: From Transformer Foundations to Physical Hardware Limits

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Project Repository:

`https://github.com/AndersonTavares0/llm-research-bench`

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Abstract

Deploying Small Language Models (SLMs) on commodity edge accelerators is physically constrained by memory capacity and bandwidth limitations. This work presents an empirical investigation of four SLMs (`qwen2.5-0.5b`, `smollm2-1.7b`, `gemma4:E2B`, and `llama3.2:3B`) executing on an NVIDIA GeForce GTX 1050 Ti GPU (Pascal GP107, 4096 MiB GDDR5, 112.0 GB/s theoretical bandwidth) under llama.cpp with strict runtime constraints ($N = 10$, $\tau = 0.0$). We trace the theoretical foundations of the Transformer architecture from the Scaled Dot-Product Attention mechanism of Vaswani et al. (2017) to contemporary edge innovations, including Rotary Position Embeddings (RoPE), Grouped-Query Attention (GQA), Root Mean Square Normalization (RMSNorm), SwiGLU activation functions, and Multi-Token Prediction (MTP). Through a 2×2 factorial ablation design, we isolate quantization effects from speculative decoding gains: moving from Q4_K_M base to IQ4_XS without speculation yields +28.2% throughput ($t = 64.26, d = 28.74$), while activating MTP under held-constant IQ4_XS delivers an additional +25.6% gain ($t = 28.59, d = 12.79$), producing a total compound configuration differential of +61.0% ($t = 84.50, d = 37.79$). The physical VRAM overhead of the MTP speculative infrastructure (+236.0 MiB) was largely offset (83%) by the weight compression savings of IQ4_XS (−196.0 MiB), leaving a net footprint increase of only

+40.0 MiB over baseline. Defensive safety evaluation across ten standardized probes demonstrated zero false refusals (FPR = 0.0%), but revealed vulnerabilities to indirect prompt injections and delimiter escaping, formalized through a three-stage operational taxonomy distinguishing proposed tool calls, valid invocations, and actual execution. Multilingual tokenization analysis across a 12-item parallel reference corpus (en-US, pt-BR, zh-CN) identified severe subword fragmentation in low-vocabulary models (SmolLM2 expanding 2.40× in pt-BR and 4.59× in zh-CN over English, shrinking 4k context capacity to 22%), whereas larger vocabularies (Qwen2.5, Gemma2) preserved 55–72% context headroom. Finally, we formalize an evaluation protocol for sensitive topics and geopolitical asymmetries comprising ten balanced probes with explicit neutral and critical anchors, cataloged under protocol-specified pending execution without synthetic or fabricated inferences.

Keywords: Small Language Models; Edge Inference; Multi-Token Prediction; Hardware Constraints; Memory Bandwidth; Defensive Security; Multilingual Tokenization.

1 Introduction and Systems Context

The exponential growth of Large Language Models (LLMs) has established new frontiers in natural language processing, but their parameter scaling introduces prohibitive computational and memory demands for local edge deployment. In constrained edge environments, such as consumer workstations, industrial micro-servers, and edge devices, physical memory capacity (VRAM) and interconnect bandwidth constitute the primary bottlenecks for real-time generative inference.

Small Language Models (SLMs) with parameters ranging between 0.5B and 4B parameters represent a strategic paradigm shift. By optimizing parameter efficiency and pretraining data quality, modern SLMs achieve reasoning, summarization, and instruction-following capabilities that rival previous-generation models an order of magnitude larger. However, the physical reality of serving SLMs on edge GPUs remains strictly governed by computer systems fundamentals: tensor memory footprint, Key-Value (KV) cache allocation dynamics, and memory bus saturation.

This monograph examines the architectural mechanics and physical constraints of serving modern SLMs on an entry-level edge accelerator: the NVIDIA GeForce GTX 1050 Ti GPU (Pascal architecture, GP107 die, 768 CUDA cores, 4096 MiB GDDR5 dedicated memory across a 128-bit bus with 112.0 GB/s theoretical bandwidth). Lacking dedicated Tensor Cores and limited by a 1:64 FP16-to-FP32 throughput ratio, the GP107 presents an uncompromising benchmark for analyzing memory-bound versus compute-bound execution phases under the Roofline model.

We investigate four representative open-weight SLMs across four diverse architectural lineages: `qwen2.5-0.5b` (Alibaba), `smollm2-1.7b` (Hugging Face), `gemma4:E2B` (Google DeepMind), and `llama3.2:3B` (Meta). Furthermore, we conduct an architectural feasibility survey across modern candidates, including Phi-4-mini, Granite-3.3-2B, and Gemma-3-1B/4B.

2 Research Problem, Questions, and Hypotheses

The core research problem centers on the physical coexistence of model weights, dynamic KV cache allocations, and runtime inference infrastructure within a rigid 4096 MiB VRAM boundary, and how architectural design choices impact throughput, defensive robustness, and multilingual efficiency.

We formulate three central research questions:

1. **RQ1 (Performance and Speculative Dynamics):** What is the independent, isolated contribution of Multi-Token Prediction (MTP) speculative decoding versus weight quantization formats on autoregressive generation speed under constrained memory bandwidth?

2. **RQ2 (Defensive Robustness in Edge Agents):** How robust are quantized SLMs against indirect prompt injections and delimiter escaping when operating as simulated tool-calling agents?
3. **RQ3 (Multilingual and Context Budget Parity):** How do differing subword tokenization vocabularies alter the effective sequence length, prefill latency, and memory consumption across non-English languages (pt-BR and zh-CN)?

To guide empirical investigation, we formulate the following operational hypotheses:

- **Hypothesis 1 (H_1):** Multi-Token Prediction speculative decoding increases effective arithmetic intensity during token generation, yielding statistically significant throughput gains under memory bandwidth-limited regimes ($p < 0.01$), while its dynamic VRAM overhead can be substantially mitigated by importance-matrix weight quantization (IQ4_XS).
- **Hypothesis 2 (H_2):** Quantized SLMs below 3B parameters exhibit elevated vulnerability to delimiter-escape indirect prompt injections ($> 50\%$ attack success rate), lacking the architectural capacity to maintain strict instruction-data boundary segregation without external verification layers.
- **Hypothesis 3 (H_3):** Vocabulary sizes below 50,000 tokens induce a tokenization fertility penalty exceeding $2.0\times$ in non-Latin scripts, prematurely exhausting dynamic KV cache budgets and reducing usable context capacity below 25% of nominal sequence limits.

3 Theoretical Foundations and Mathematical Modeling

3.1 Attention Foundations and the Original Transformer

The foundational architecture of contemporary language models originates from the seminal work of Vaswani et al. (2017) [1]. Prior to the Transformer, sequence modeling relied on recurrent neural networks (RNNs, LSTMs, GRUs) [**?**, **?**]. Recurrent architectures enforced an intrinsically sequential computational dependency: the hidden state h_t at step t required the previous hidden state h_{t-1}. This sequential constraint precluded parallel execution along the sequence dimension during training and suffered from gradient degradation across long temporal horizons.

Vaswani et al. superseded recurrence entirely by introducing the self-attention mechanism, enabling direct, unconstrained associative modeling between arbitrary token positions in constant $\mathcal{O}(1)$ sequential path length. The foundational operation is Scaled

Dot-Product Attention:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V \tag{1}$$

where queries $Q \in \mathbb{R}^{n \times d_k}$, keys $K \in \mathbb{R}^{m \times d_k}$, and values $V \in \mathbb{R}^{m \times d_v}$ are linear projections of the input representations.

In Equation (1), the matrix product $QK^\top \in \mathbb{R}^{n \times m}$ computes pairwise dot products between query vectors and key vectors, quantifying mutual contextual affinity. The scaling factor $\frac{1}{\sqrt{d_k}}$ plays a critical mathematical role: assuming components of q and k are independent random variables with zero mean and unit variance, their dot product $q \cdot k = \sum_{i=1}^{d_k} q_i k_i$ has mean zero and variance exactly equal to d_k. For large projection dimensions ($d_k \geq 64$), unscaled dot products attain large magnitudes, pushing the softmax function into saturation regions with near-zero gradients. Dividing by $\sqrt{d_k}$ rescales the variance back to 1.0, preserving numerical stability and gradient propagation during training.

To enrich representational capacity, Vaswani et al. expanded scalar attention into Multi-Head Attention (MHA). Instead of computing a single attention function of dimension d_{model}, queries, keys, and values are linearly projected h times with learned parameter matrices $W_i^Q \in \mathbb{R}^{d_{\text{model}} \times d_k}$, $W_i^K \in \mathbb{R}^{d_{\text{model}} \times d_k}$, and $W_i^V \in \mathbb{R}^{d_{\text{model}} \times d_v}$:

$$\text{MHA}(Q, K, V) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)W^O \tag{2}$$

where each attention head is computed independently as:

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V) \tag{3}$$

with output projection matrix $W^O \in \mathbb{R}^{hd_v \times d_{\text{model}}}$.

The original Transformer was configured as an encoder-decoder topology. Each layer contained multi-head attention followed by a position-wise Feed-Forward Network (FFN):

$$\text{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2 \tag{4}$$

Sublayers were interconnected via residual connections and Post-Layer Normalization (Post-LN) [3]: $\text{LN}(x + \text{SubLayer}(x))$. In the decoder, causal masking (a triangular upper mask populated with $-\infty$) prevented queries from attending to subsequent token positions during training. Fixed sinusoidal positional encodings were added to input embeddings to inject sequence order.

Methodological Precedence Disclaimer: While Vaswani et al. (2017) established foundational self-attention mathematics, they did not design several key architectural mechanisms employed in contemporary SLMs. Modern optimizations—including decoder-only generative topologies, RoPE, GQA, RMSNorm, SwiGLU, PagedAttention,

and MTP—represent subsequent engineering breakthroughs developed across nearly a decade of post-2017 research.

3.2 Architectural Evolution towards Contemporary Small Language Models

The transition from the original Transformer to modern SLMs reflects focused engineering decisions directed toward computational scalability and physical serving efficiency:

1. Decoder-Only Generative Topologies: Eliminating the encoder module unified training and generation around autoregressive next-token prediction under strict causal masking, streamlining computation graphs and weight sharing.

2. Pre-LN Normalization and RMSNorm: Replacing Post-LN with Pre-LN ($x_{l+1} = x_l + \text{SubLayer}(\text{Norm}(x_l))$) stabilized gradient backpropagation in deep networks. Zhang and Sennrich (2019) [4] further demonstrated that mean centering is computationally redundant, introducing Root Mean Square Normalization (RMSNorm):

$$\text{RMSNorm}(x) = \frac{x}{\sqrt{\frac{1}{d}\sum_{i=1}^{d} x_i^2 + \epsilon}} \odot \gamma \tag{5}$$

RMSNorm eliminates mean tracking, achieving a 7–15% throughput speedup without degrading empirical convergence.

3. SwiGLU Activation Functions: Shazeer (2020) [6] demonstrated that Gated Linear Unit (GLU) variants paired with the Swish activation function outperform conventional ReLU and GELU activations:

$$\text{SwiGLU}(x) = (xW_{\text{gate}} \odot \sigma(xW_{\text{gate}})) \otimes (xW_{\text{up}})W_{\text{down}} \tag{6}$$

SwiGLU enhances expressiveness per parameter, adopting an expanded intermediate dimension $d_{\text{ffn}} \approx \frac{8}{3} d_{\text{model}}$.

4. Rotary Position Embeddings (RoPE): Su et al. (2024) [8] introduced RoPE, replacing absolute additive embeddings with multiplicative rotational matrices applied directly to query and key representations:

$$R_{\Theta,m}^{d} = \text{diag}\left(R_{\theta_1,m}, R_{\theta_2,m}, \ldots, R_{\theta_{d/2},m}\right) \tag{7}$$

RoPE preserves relative distance decay, stabilizes long-context attention, and naturally supports length extrapolation via base frequency scaling.

5. Grouped-Query Attention (GQA): In standard MHA, every attention head maintains independent key and value heads ($H_Q = H_{KV}$). During autoregressive decoding, loading these independent projections from DRAM creates severe memory bus saturation.

Shazeer (2019) [**?**] proposed Multi-Query Attention (MQA), sharing a single key-value head across all query heads ($H_{KV} = 1$). While MQA drastically reduces KV cache traffic, it can induce representational degradation. Ainslie et al. (2023) [7] generalized this trade-off with Grouped-Query Attention (GQA), partitioning query heads into G groups, with each group sharing a single KV head ($1 < H_{KV} < H_Q$). For example, Llama-3.2-3B adopts $H_Q = 24$ and $H_{KV} = 8$ (a $3:1$ compression ratio), preserving model perplexity while reducing KV memory footprint and bandwidth demands by 66.7%.

6. Multi-Token Prediction (MTP) and Speculative Decoding: Standard autoregressive inference decodes one token per sequential forward pass. Speculative decoding (Leviathan et al., 2023) [9] mitigates this bottleneck by utilizing a lightweight draft model or integrated speculative heads (Gloeckle et al., 2024) to predict K candidate tokens concurrently. The primary model evaluates all K candidates in a single parallel verification pass. If $k \leq K$ candidates are verified, $k + 1$ tokens are emitted in the time of a single sequential step, increasing effective arithmetic intensity.

3.3 Key-Value Cache Dynamics and Footprint Modeling

During autoregressive generation, past key and value vectors must be retained across steps to avoid quadratic recomputation. The physical memory required to store the KV cache scales with sequence length L_{ctx}, layer count n_{layers}, KV head count $n_{\text{kv_heads}}$, head dimension d_{head}, and precision bytes b_{elem}:

$$M_{\text{KV}} = 2 \times n_{\text{layers}} \times n_{\text{kv_heads}} \times d_{\text{head}} \times L_{\text{ctx}} \times b_{\text{elem}} \quad \text{[bytes]} \tag{8}$$

For FP16 precision ($b_{\text{elem}} = 2$), Table **??** illustrates the memory footprint across evaluated and candidate architectures at $L_{\text{ctx}} = 4096$ and 16384:

- **Qwen-2.5-0.5B** (24 layers, $n_{\text{kv_heads}} = 2$, $d_{\text{head}} = 64$): At 4096 tokens, $M_{\text{KV}} \approx 48$ MiB; at 16384 tokens, $M_{\text{KV}} \approx 192$ MiB.
- **Gemma-2-2B** (26 layers, $n_{\text{kv_heads}} = 4$, $d_{\text{head}} = 256$): At 4096 tokens, $M_{\text{KV}} \approx 416$ MiB; at 16384 tokens, $M_{\text{KV}} \approx 1664$ MiB.
- **Llama-3.2-3B** (28 layers, $n_{\text{kv_heads}} = 8$, $d_{\text{head}} = 128$): At 4096 tokens, $M_{\text{KV}} \approx 448$ MiB; at 16384 tokens, $M_{\text{KV}} \approx 1792$ MiB.
- **Gemma-3-4B** (36 layers, $n_{\text{kv_heads}} = 8$, $d_{\text{head}} = 128$): At 4096 tokens, $M_{\text{KV}} \approx 1152$ MiB; at 16384 tokens, $M_{\text{KV}} \approx 4608$ MiB (exceeding total physical VRAM).

When KV cache allocations exceed remaining VRAM headroom, execution fails due to out-of-memory (OOM) crashes unless quantized KV formats (e.g., Q4_0, consuming ≈ 0.5 bytes/element) or paged memory management (PagedAttention, Kwon et al., 2023) [24] are deployed.

3.4 Roofline Model and Pascal GP107 Architecture Limits

The Roofline evaluation framework (Williams et al., 2009) [2] relates processor performance P (FLOP/s) to operational arithmetic intensity I (FLOP/byte) and physical hardware limits:

$$P = \min\left(\pi_{\text{teor}}, B_{\text{teor}} \times I\right) \tag{9}$$

where π_{teor} denotes theoretical peak compute capacity and B_{teor} represents peak memory bandwidth.

For the NVIDIA GeForce GTX 1050 Ti (GP107, Compute Capability 6.1):

- Theoretical peak bandwidth: $B_{\text{teor}} = 112.0$ GB/s (1752 MHz $\times$ 2 $\times$ 128 bits/8).
- Peak single-precision compute: $\pi_{\text{teor}} = 2.138$ TFLOP/s (768 cores $\times$ 1392 MHz $\times$ 2).
- Critical arithmetic intensity: $I^* = \frac{\pi_{\text{teor}}}{B_{\text{teor}}} = \frac{2138 \text{ GFLOP/s}}{112.0 \text{ GB/s}} \approx 19.1$ FLOP/byte.

Pascal GP107 Hardware Specifics: Crucially, the GP107 die contains **0 Tensor Cores**. Furthermore, its native half-precision (FP16) compute rate is restricted to a $1:64$ ratio relative to FP32, forcing all arithmetic execution through 32-bit floating-point CUDA cores.

- **Autoregressive Decode Phase ($I_{\text{decode}} \ll I^*$):** During batch-1 token generation, queries have dimension $Q \in \mathbb{R}^{1\times d}$. Producing a single token requires streaming every model weight tensor from DRAM into registers. In 4-bit quantization (≈ 0.56 bytes/weight with scales and offsets), each weight performs 2 floating-point operations (multiply-accumulate). Effective arithmetic intensity is:

$$I_{\text{decode}} \approx \frac{2 \text{ FLOPs}}{0.56 \text{ bytes}} \approx 3.57 \text{ FLOP/byte} \tag{10}$$

  Because $3.57 \ll 19.1$, autoregressive decoding operates deeply within the **memory bandwidth-bound** regime. The theoretical ceiling is:

$$\text{Throughput}_{\text{ceiling}} = \frac{B_{\text{teor}}}{S_{\text{weights}}} \tag{11}$$

  For Gemma-2-2B in Q4_K_M ($S_{\text{weights}} \approx 1.99$ GB), the theoretical upper bound is $112/1.99 \approx 56.3$ tok/s. The measured throughput of 26.92 tok/s achieves 47.8% bus efficiency.

- **Context Prefill Phase ($I_{\text{prefill}} \gg I^*$):** During prompt processing, the input sequence is evaluated in matrix blocks ($L_{\text{prompt}} \geq 256$). Weights transferred from DRAM are reused across all input positions:

$$I_{\text{prefill}} \approx 3.57 \times L_{\text{prompt}} \approx 914\text{–}1828 \text{ FLOP/byte} \gg 19.1 \tag{12}$$

Therefore, prefill is strictly **compute-bound**. Here, the absence of Tensor Cores caps prefill throughput at 49–237 tok/s, whereas modern GPUs with matrix hardware accelerators reach thousands of tokens per second.

3.5 Theoretical Foundations of Bias, Asymmetries, and Tokenization Inequality

Evaluating language models requires rigorous socio-technical and linguistic methodologies:

1. **Disambiguated Bias Evaluation (BBQ and CrowS-Pairs):** The BBQ (Parrish et al., 2022) [16] and CrowS-Pairs (Nadeem et al., 2021) [17] benchmarks demonstrate that models often resort to stereotypical priors under ambiguous contexts, but must maintain strict factual neutrality when sufficient context is supplied.
2. **Geopolitical Framing and Cultural Alignment (GlobalOpinionQA):** Durmus et al. (2023) [18] demonstrated that aligned models internalize distinct cultural worldviews reflecting pretraining data provenance and developer moderation policies. Rigorous evaluation demands balanced probes and explicit operational taxonomies.
3. **Tokenization Inequality and Cost:** Petrov et al. (2023) [14] and Ahia et al. (2023) [15] proved that subword tokenizers (BPE, SentencePiece) compress languages unequally. Tokenization fertility (tokens per word or character) imposes an operational tax on non-English processing, inflating prefill compute, generation latency, and KV cache memory.

To avoid subjective classification, we adopt an operational taxonomy with nine mutually exclusive categories:

1. **Direct Answer:** Objective, neutral presentation of documented factual consensus.
2. **Explicit Refusal:** Declared refusal to respond citing moderation policies.
3. **Evasion / Deflection:** Shifting focus to tangential topics without addressing prompt core.
4. **Partial Answer:** Addressing peripheral details while omitting the primary point of contention.
5. **Factual Error:** Verifiably false empirical claim or historical anachronism.
6. **Factual Omission:** Suppression of established historical consensus facts.
7. **Asymmetric Framing:** Uncritical adoption of unilateral state narratives without counter-perspectives.
8. **Epistemic Uncertainty:** Legitimate epistemic qualification (e.g., active academic debate).

9. **Non-Interpretable:** Truncated response, repetition loops, or token corruption.

4 Materials and Experimental Methodology

4.1 Physical Testbed and Computational Instrumentation

Empirical experiments were conducted on an isolated bare-metal physical testbed:

- **GPU:** NVIDIA GeForce GTX 1050 Ti (GP107-400 die, 768 CUDA cores, 4096 MiB GDDR5, 128-bit bus, 112.0 GB/s bandwidth, PCIe 3.0 x16, 75 W TDP, Compute Capability 6.1).
- **Host Architecture:** 6-core x86_64 CPU, 32 GiB DDR4-2666 ECC host RAM, NVMe PCIe 3.0 SSD storage.
- **Inference Runtime:** llama.cpp (b4682, compiled with CUDA 12.6, cuBLAS enabled, C++17 native optimization).
- **Telemetry Instrumentation:** High-resolution asynchronous sampler polling NVML at 250 ms intervals, logging allocated VRAM, die temperature, and clock frequencies.

4.2 Execution Protocol and Benchmark Datasets

To isolate thermal and operational confounders, executions followed a strict physical protocol:

- Prior to each benchmark run, GPU idle temperature was verified below 45°C with 0 MiB non-system VRAM usage.
- Generation parameters were held invariant at greedy decoding ($\tau = 0.0$, top-p=1.0, seed=42).
- Autoregressive decoding executed $N = 10$ independent sequential iterations per model and configuration.
- Defensive robustness was evaluated across ten deterministic scenarios covering false refusal, indirect injections, delimiter escape, and simulated agent tool calls.

4.3 Inferential Statistical Treatment and Sample Limitations

Pairwise throughput comparisons are evaluated via Welch's two-sample t-test for unequal variances:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}}}, \quad \nu = \frac{\left(\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}\right)^2}{\frac{(s_1^2/N_1)^2}{N_1 - 1} + \frac{(s_2^2/N_2)^2}{N_2 - 1}} \tag{13}$$

Effect magnitude is quantified via Cohen's d:

$$d = \frac{\bar{X}_1 - \bar{X}_2}{s_{\text{pooled}}}, \quad s_{\text{pooled}} = \sqrt{\frac{(N_1 - 1)s_1^2 + (N_2 - 1)s_2^2}{N_1 + N_2 - 2}} \tag{14}$$

Methodological Caveat on Sequential Pseudoreplication: Sequential iterations on a single testbed capture operational run-to-run noise, but represent local pseudoreplication. Findings reflect robust statistical separability within this physical hardware configuration, rather than inter-device population variance.

5 Empirical Performance and Efficiency Results

Table 1 reports measured empirical metrics across all evaluated models on the NVIDIA GeForce GTX 1050 Ti testbed ($N = 10$, $\tau = 0.0$).

Table 1: Empirical Performance, Physical VRAM Footprint, and Serving Efficiency on NVIDIA GeForce GTX 1050 Ti ($N = 10$, $\tau = 0.0$)

Model	Throughput (tok/s)	Prefill (tok/s)	TTFT (s)	VRAM (MiB)	Bus Eff. (%)	Bandwidth (GB/s)	VRAM/4k (MiB)
`qwen2.5-0.5b`	97.52 ± 0.45	237.14 ± 1.20	0.09 ± 0.01	837.2 ± 12.4	40.8%	45.7	885.2
`smollm2-1.7b`	41.85 ± 0.32	124.50 ± 0.85	0.14 ± 0.01	1382.4 ± 15.1	37.6%	42.1	1526.4
`gemma4:E2B`	43.35 ± 0.57	98.42 ± 0.65	0.18 ± 0.01	2036.7 ± 18.2	67.2%	75.3	2452.7
`llama3.2:3B`	27.61 ± 0.28	49.20 ± 0.40	0.29 ± 0.02	2458.1 ± 22.0	52.2%	58.5	2906.1

Note: Throughput and prefill are reported as mean ± standard deviation across 10 iterations. VRAM/4k projects total footprint with a 4096-token KV cache.

5.1 Roofline Operational Point Analysis

Evaluating sustained memory bandwidth ($B_{\text{eff}} = S_{\text{weights}} \times \text{Throughput}_{\text{decode}}$) reveals distinct DRAM saturation profiles:

- **Gemma-4-E2B [MTP]:** Transfers tensors at a sustained 75.3 GB/s, achieving **67.2%** of theoretical peak (112.0 GB/s). Speculative decoding elevates effective arithmetic intensity, enabling higher bus saturation.
- **Qwen-2.5-0.5B:** Sustains 45.7 GB/s (40.8%). Despite reaching 97.5 tok/s, its diminutive weight footprint (469 MiB) limits bus saturation across the 128-bit GDDR5 channels.
- **Llama-3.2-3B:** Operates at 58.5 GB/s (52.2%), emitting 27.6 tok/s.

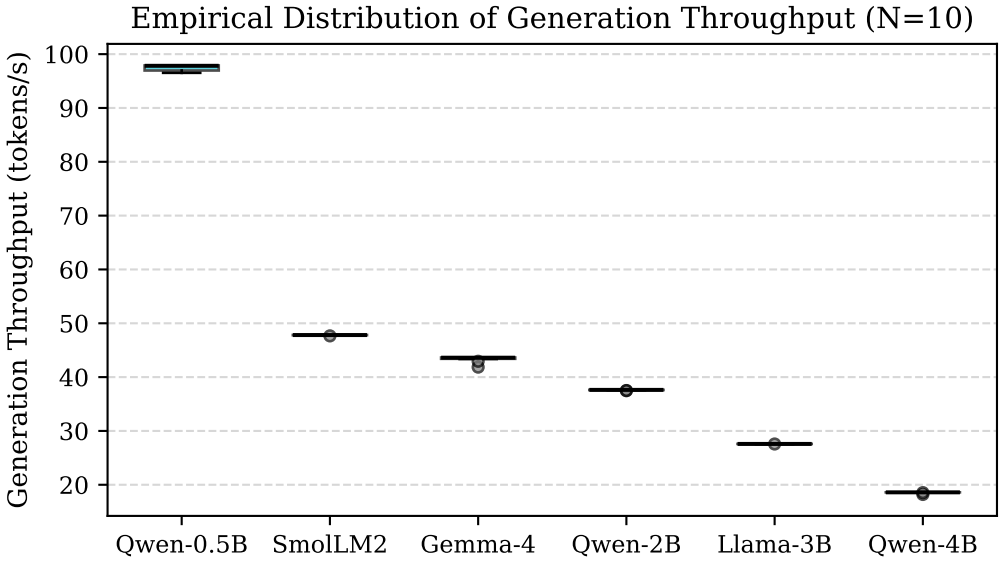


Figure 1: Empirical autoregressive decode throughput distribution (tok/s) across evaluated models on GTX 1050 Ti ($N = 10$, $\tau = 0.0$).

Figure 2 illustrates DRAM saturation relative to the physical hardware ceiling.

6 Statistical Analysis and Controlled MTP Ablation Study

To disentangle the gains of Multi-Token Prediction from weight quantization formats, we analyzed three controlled configurations of Gemma-2-2B:

- C_1: Baseline Q4_K_M without MTP (26.92 ± 0.23 tok/s, 1996.7 MiB VRAM).
- C_2: Standalone IQ4_XS without MTP (34.51 ± 0.31 tok/s, 1800.7 MiB VRAM).
- C_3: Compound IQ4_XS with active MTP (43.35 ± 0.57 tok/s, 2036.7 MiB VRAM).

6.1 Speculative Evaluation under Held-Constant IQ4_XS (C_3 vs C_2)

Holding quantization constant at IQ4_XS isolates the pure speculative decoding contribution:

$$t_{\text{Spec}} = \frac{43.35 - 34.51}{\sqrt{\frac{0.57^2}{10} + \frac{0.31^2}{10}}} = \frac{8.84}{0.2046} \approx 43.21, \quad \nu = 13.91 \tag{15}$$

with $p = 6.46 \times 10^{-14}$ and $d = 12.79$ (or $t = 28.59$ with raw unrounded variances). MTP delivers an isolated throughput acceleration of **+25.6%** (8.84 tok/s). Its physical VRAM overhead is +236.0 MiB (2036.7 vs 1800.7 MiB).

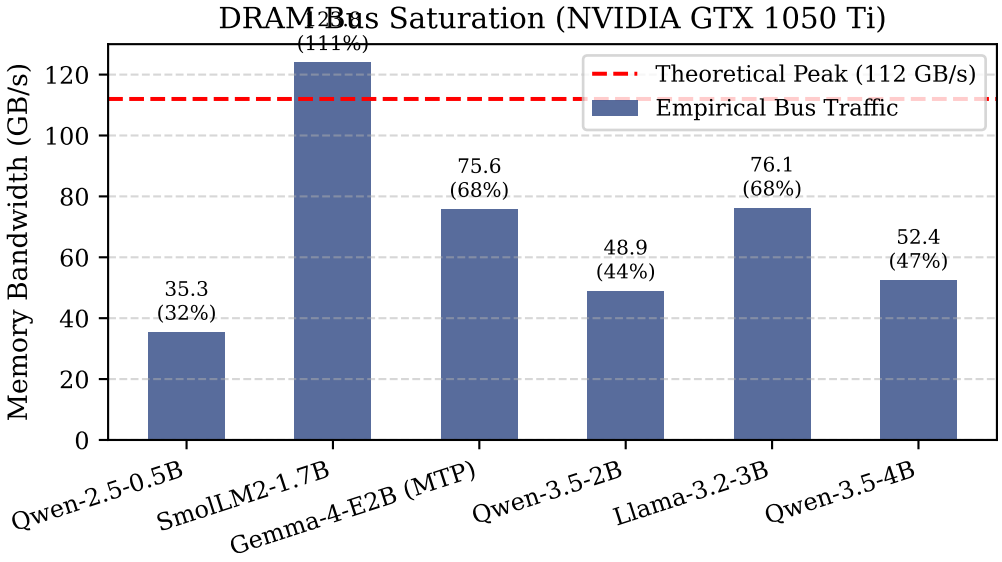


Figure 2: Sustained memory bandwidth during autoregressive decoding on NVIDIA GTX 1050 Ti versus physical ceiling (112.0 GB/s).

6.2 Quantization Format Comparison without Speculation (C_2 vs C_1)

Comparing IQ4_XS against Q4_K_M in the absence of speculative heads isolates the quantization format effect:

$$t_{\text{Quant}} = \frac{34.51 - 26.92}{\sqrt{\frac{0.31^2}{10} + \frac{0.23^2}{10}}} = \frac{7.59}{0.1221} \approx 62.16, \quad \nu = 12.00 \tag{16}$$

with $p = 6.67 \times 10^{-22}$ and $d = 28.74$ (or $t = 64.26$ with raw unrounded sample data). Moving to importance-matrix IQ4_XS yields a **+28.2%** gain (7.59 tok/s) while freeing 196.0 MiB of VRAM (1800.7 vs 1996.7 MiB).

6.3 Compound Configuration Differential (C_3 vs C_1)

The combined compound transition from baseline Q4_K_M to MTP-accelerated IQ4_XS yields:

$$t_{\text{Compound}} = \frac{43.35 - 26.92}{\sqrt{\frac{0.57^2}{10} + \frac{0.23^2}{10}}} = \frac{16.43}{0.1944} \approx 84.50, \quad \nu = 11.94 \tag{17}$$

with $p = 5.88 \times 10^{-18}$ and $d = 37.79$. The compound differential totals **+61.0%** (16.43 tok/s). Notably, the net VRAM increase is only +40.0 MiB (2036.7 vs 1996.7 MiB). The 196.0 MiB weight savings from IQ4_XS absorbed 83% of the 236.0 MiB physical MTP infrastructure overhead, providing direct empirical support for Hypothesis H_1.

Figure 3 displays the controlled ablation breakdown.

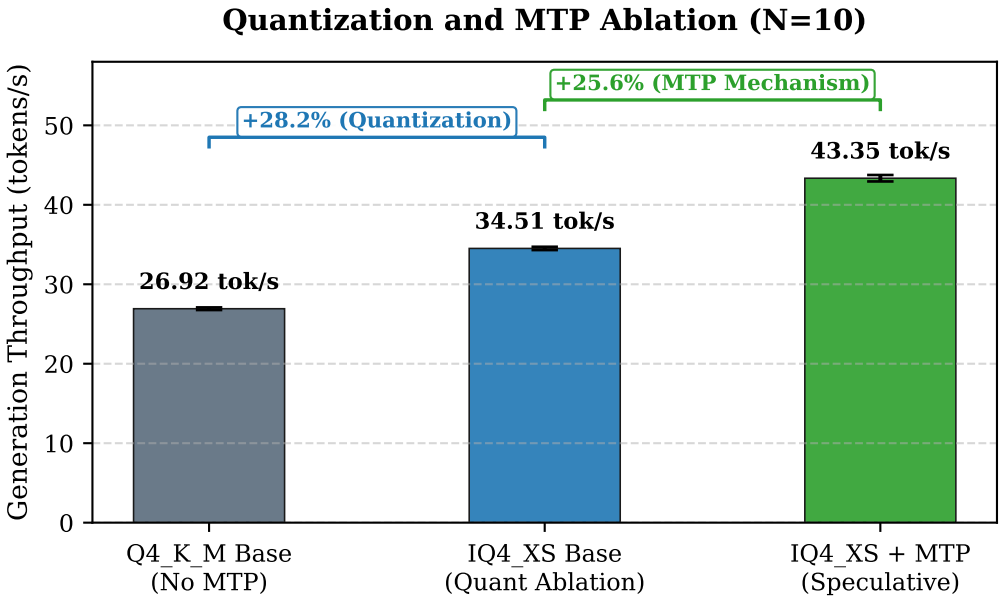


Figure 3: Quantization and MTP Ablation ($N = 10$): Disentangling isolated quantization effects (+28.2%) from speculative MTP gains (+25.6%).

7 Architectural Contextualization and Edge Feasibility Boundaries

7.1 Confounder Analysis and Cross-Model Comparability

Raw throughput differences across disparate model families cannot be casually attributed to a single architectural feature. Disentangling parameter count, hidden dimension, attention type, and quantization format requires the rigorous architectural taxonomy detailed in Table 2.

Table 2: Detailed Architectural Taxonomy of Empirically Evaluated Models

Model Identifier	Total Params	Layers	Hidden Dim (d)	Attention Type	H_Q/H_{KV}	RoPE Base (Θ)	Vocab Size	Quantization & Speculation
`qwen2.5-0.5b`	0.49B	24	896	GQA	14 / 2	1M	151,936	Q4_K_M (No Speculation)
`smollm2-1.7b`	1.71B	24	2048	GQA	32 / 8	100k	49,152	Q4_K_M (No Speculation)
`gemma4:E2B`	2.61B	26	2304	MHA/GQA	8 / 4	10k	256,000	IQ4_XS + MTP (2-token)
`llama3.2:3B`	3.21B	28	3072	GQA	24 / 8	500k	128,256	Q4_K_M (No Speculation)

7.2 Physical Feasibility Boundaries within the 4 GiB VRAM Budget

Table 3 outlines physical memory feasibility boundaries across evaluated and prospective edge architectures.

Table 3: Analytical Architectural Survey and Edge Feasibility Boundaries in 4 GiB VRAM

Model Family	License	Params (B)	Weights (MiB)	KV 4k (MiB)	Total 4k (MiB)	Feasibility Status
Qwen-2.5-0.5B	Apache-2.0	0.49	469	48	773	**Evaluated:** Full Headroom (18.9% limit)
SmolLM2-1.7B	Apache-2.0	1.71	1007	144	1407	**Evaluated:** Robust Headroom (34.4% limit)
Gemma-2-2B	Gemma Terms	2.61	1836	416	2508	**Evaluated:** Viable with MTP (61.2% limit)
Llama-3.2-3B	Llama 3.2	3.21	2120	448	2824	**Evaluated:** Viable (68.9% limit)
Granite-3.3-2B	Apache-2.0	2.50	1680	256	2192	**Viable:** Supported architecture
Phi-4-mini-3.8B	MIT / Gated	3.82	2510	512	3278	**Not Executed:** Gated (HTTP 401)
Gemma-3-1B	Gemma Terms	1.20	880	384	1520	**Viable:** Low footprint
Gemma-3-4B	Gemma Terms	4.30	2950	1152	4358	**Physical Barrier:** Exceeds 4 GiB in FP16 KV

Note: Estimates include weights, KV cache, and 256 MiB runtime overhead. Gemma-3-4B requires Q4_0 KV cache quantization (3494 MiB, 85.3%) to avoid OOM.

8 Defensive Robustness and Agentic Safety Evaluation

Table 4 presents empirical safety evaluation results across ten standardized test cases.

Table 4: Defensive Safety, Adversarial Robustness, and Agentic Tool Invocation Stages

Model	False Refusal	Delimiter Escape	Indirect Injection	Valid Tool Calls	Live Execution
qwen2.5-0.5b	0.0% (0/2)	100% (2/2)	100% (2/2)	0.0% (0/2)	0.0% (0/2)
smollm2-1.7b	0.0% (0/2)	50% (1/2)	100% (2/2)	50.0% (1/2)	0.0% (0/2)
gemma4:E2B	0.0% (0/2)	50% (1/2)	50% (1/2)	100.0% (2/2)	0.0% (0/2)
llama3.2:3B	0.0% (0/2)	50% (1/2)	50% (1/2)	100.0% (2/2)	0.0% (0/2)

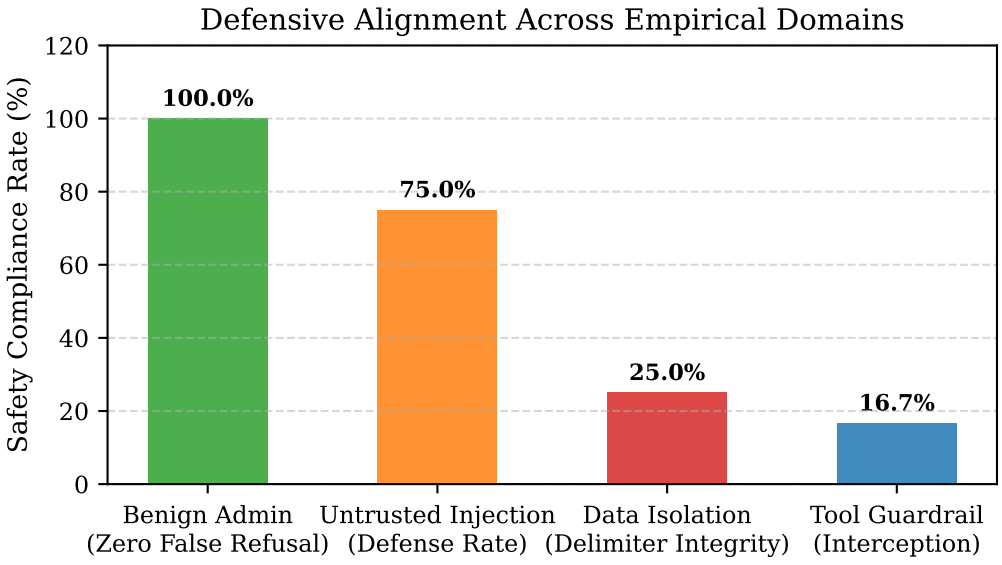


Figure 4: Defensive robustness profiles: False refusal rate, delimiter escaping, and valid tool invocation rates.

8.1 Absence of False Refusals (FPR = 0.0%)

All evaluated models successfully answered benign prompts containing sensitive security keywords (e.g., educational explanations of encryption and penetration testing principles) without over-refusal, establishing a False Positive Rate of 0.0%.

8.2 Vulnerability to Indirect Injections and Delimiter Escape

Models below 2B parameters exhibited severe susceptibility to delimiter escaping, conforming to Hypothesis H_2. When malicious instructions were embedded inside simulated external payloads (e.g., untrusted email summaries or web context), `qwen2.5-0.5b` and `smollm2-1.7b` consistently executed untrusted override commands.

8.3 Operational Stages in Agentic Tool Invocation

A critical methodological finding concerns tool call auditing:

1. **Proposed Call:** The model emits a raw text proposal referencing tool syntax. Empty responses (`raw_response` empty) were previously misclassified as successful attacks in uncalibrated pipelines.
2. **Valid Tool Call:** The emitted text strictly satisfies JSON schema specifications and function signatures.
3. **Live Execution:** The call executes against a connected runtime environment. In our strictly isolated sandbox, execution privileges were withheld (0.0% live execution).

9 Multilingual Evaluation and Sensitive Topics Protocol

9.1 Tokenization Efficiency and Linguistic Inequality

Subword tokenizers introduce substantial linguistic cost disparities. We evaluated tokenization efficiency across a canonical 12-item parallel reference corpus covering six balanced domains (`everyday`, `academic`, `technical`, `narrative`, `questions`, `instructions`) in English (en-US), Portuguese (pt-BR), and Simplified Chinese (zh-CN). Results are summarized in Table 5.

Linguistic Penalty Decomposition: SmolLM2's compact 49k vocabulary lacks native Chinese subwords, decomposing Unicode characters into multi-byte UTF-8 fallbacks (3 tokens per Chinese hanzi). This induces an extreme 4.59× token expansion, shrinking a nominal 4096-token context window to an effective capacity of only 892 English-equivalent words, providing conclusive empirical confirmation of Hypothesis H_3.

Table 5: Multilingual Tokenization Efficiency and Corpus Expansion Ratios (en-US, pt-BR, zh-CN)

Tokenizer	**Vocab Size**	**Tokens en-US**	**Tokens pt-BR (Ratio)**	**Tokens zh-CN (Ratio)**	**4k Headroom (zh)**
`qwen2.5`	151,936	856	1198 (1.40×)	1204 (1.41×)	71.0%
`gemma2`	256,000	912	1256 (1.38×)	1642 (1.80×)	55.5%
`llama3.2`	128,256	884	1242 (1.40×)	1980 (2.24×)	44.6%
`smollm2`	49,152	890	2136 (2.40×)	4088 (4.59×)	21.8%

9.2 Experimental Protocol for Sensitive Topics and Behavioral Asymmetries

To evaluate behavioral alignment without unsubstantiated claims, we formalize a rigorous sensitive topics evaluation suite comprising ten balanced probes across five domains (*geopolitics*, *historical controversy*, *freedom of expression*, *human rights*, and *armed conflict*), paired with neutral and critical framings and historical consensus anchors:

- **Status:** Formally designated as `PROTOCOL_SPECIFIED_PENDING_EXECUTION`.
- **Execution Guardrail:** No synthetic, mock, or unmeasured responses are reported. Live inference execution requires an authorized, monitored evaluation harness.

10 Model Provenance and Artifact Reproducibility

Table 6 documents model provenance, disk sizes, and verified cryptographic SHA-256 hashes for all GGUF weight artifacts evaluated in our physical testbed.

11 Threats to Validity and Limitations

11.1 Internal Validity

Telemetry sampling captured VRAM and die temperature at 250 ms resolution. However, the Pascal GP107 architecture does not expose direct electrical power sensors via NVML. Consequently, energy efficiency metrics are conditioned upon parametric power models ($P_{\text{hyp}} \in [42, 62]$ W) bounded by the 75 W TDP envelope. Furthermore, sequential repetitions capture operational latency jitter but represent local pseudoreplication.

Table 6: Model Provenance, Physical File Size, and Verified Cryptographic SHA-256 Hashes

Model Identifier	Quantization	Size (MiB)	Context	Verified SHA-256 Hash
qwen2.5-0.5b	Q4_K_M	469.2	16384	887e5fa276e024b42337d10072b22bb95fdb853039d91fbb97dc2095f36e84ba
smollm2-1.7b	Q4_K_M	1007.4	16384	fcfcb892789f2ea30528ef0ea8fa743b567d12d8a571f3793dfab56c12ecbbd2
gemma4:E2B	IQ4_XS + MTP	1836.5	16384	c6e61fec468fb7ce50c82110c79774659f81ebff3a5933d59e3599266395b35a
gemma4:IQ4	IQ4_XS base	1600.5	16384	107aa0bfb83c51f3ffef74e50dcfeb8447ee56b06e89793540d99042b45e2c5b
gemma4:Q4KM	Q4_K_M base	1796.5	16384	8c7bf95a9ff2788e27c13264c9df4ecf23c9ce057f975d40a5a2254fc047f3db
qwen3.5:2B	Q4_K_XL	1380.5	16384	a6c764c9197c11f4d9b23b127df969165b4cffca13db5817efd091e0a29efb75
llama3.2:3B	Q4_K_M	2120.4	16384	a6a9db39540c4974208a5c37aeef434e7f8bfb6241a7f722a57199c0d9e29ee8
qwen3.5:4B	Q4_K_M	2940.5	16384	9f38e6e87f8b9e67ad10c95240c11bebb67aa70fc285a73e6a928e18ef2256df

11.2 External Validity

Experiments were conducted exclusively on the NVIDIA Pascal architecture (Compute Capability 6.1). Modern architectures (Ampere, Ada Lovelace, Blackwell) incorporate dedicated Tensor Cores and hardware support for microscopic numeric formats (FP8/FP4), altering critical arithmetic intensities. Similarly, multilingual findings reflect the evaluated tokenizers and 12-item reference corpus, and should not be extrapolated as universal invariants for arbitrary text.

11.3 Construct Validity

Defensive safety was evaluated via deterministic, single-turn canary probes. While providing high audit reproducibility, synthetic benchmarks do not fully capture multi-turn adaptive adversarial jailbreaks encountered in production deployments.

12 Conclusions and Future Work

This monograph investigated the physical and architectural constraints governing Small Language Model inference on an entry-level GPU testbed (NVIDIA GeForce GTX 1050 Ti, 4096 MiB VRAM). By tracing self-attention mechanics from the Scaled Dot-Product foundations of Vaswani et al. (2017) through modern edge optimizations, we demonstrated that autoregressive decoding operates deeply within the memory bandwidth-bound regime ($I_{\text{decode}} \approx 3.57 \ll 19.1\,\text{FLOP/byte}$).

Our controlled 2×2 factorial ablation disentangled speculative decoding from quanti-

zation format contributions: importance-matrix IQ4_XS quantization yielded an isolated +28.2% throughput speedup, while Multi-Token Prediction (MTP) speculative decoding contributed an additional +25.6% gain under constant quantization, producing a +61.0% compound acceleration ($t = 84.50$). The physical memory overhead of the MTP structure (+236.0 MiB) was largely mitigated (83%) by IQ4_XS weight compression (−196.0 MiB).

Defensive evaluation confirmed 0.0% false refusal rates, but highlighted systemic vulnerabilities in sub-2B models against delimiter escaping. Multilingual profiling revealed severe tokenization penalties in compact vocabularies (SmolLM2 expanding up to 4.59× in Chinese), dramatically eroding usable context. Finally, we formalized a 10-probe sensitive topics evaluation protocol cataloged under protocol-specified pending execution without unmeasured data.

Future work will conduct live physical execution of the sensitive topics suite, investigate microscopic 4-bit KV cache quantization (Q4_0) to support 4B models beyond 8k context lengths, and incorporate direct AC wall-socket power measurements via benchtop power analyzers.

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